

PROTOTYPE
AUTHENTICATION & SECURITY MODULE

Smart Door Automation System for CICS (SDASFC)

Administrator Portal Login

System Operational Credentials

Username / Admin ID:

admin

Password:

.....

Security Lockout Status:

Active Protection: 30s lockout after 5 failed attempts

GUARD ACTIVE

☒ Remember session on this browser

[Forgot Password?](#)

Log In



Forgot Password & Security Challenge Verification

Step 1: Account Lookup

Step 2: Security Question

Step 3: Password Reset

Admin Username:

admin

VERIFIED

Registered Security Question:

What is the primary server room designation?

Security Answer:

CICS-LAB-402

New Password:

.....

Confirm New Password:

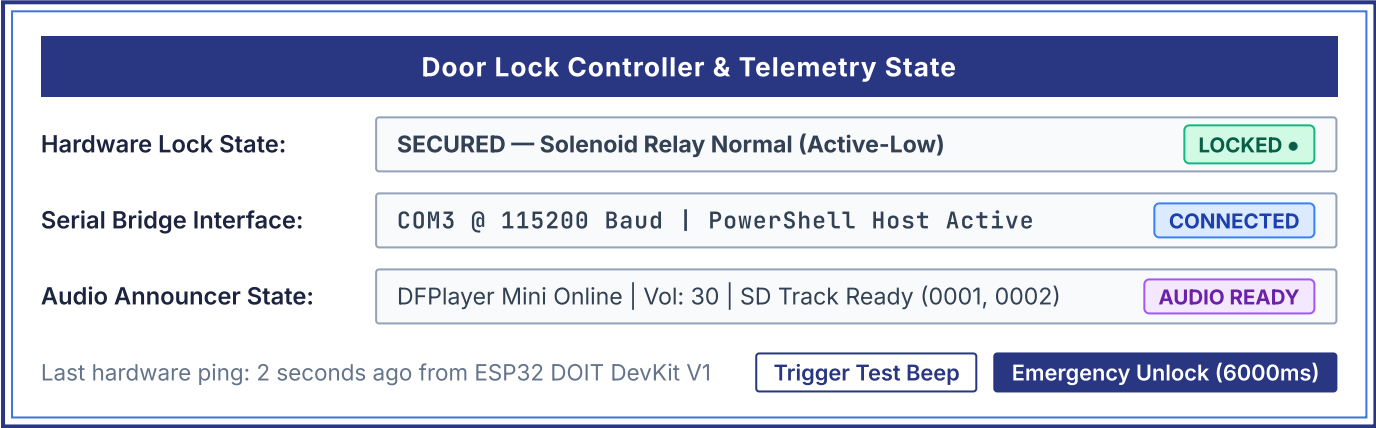
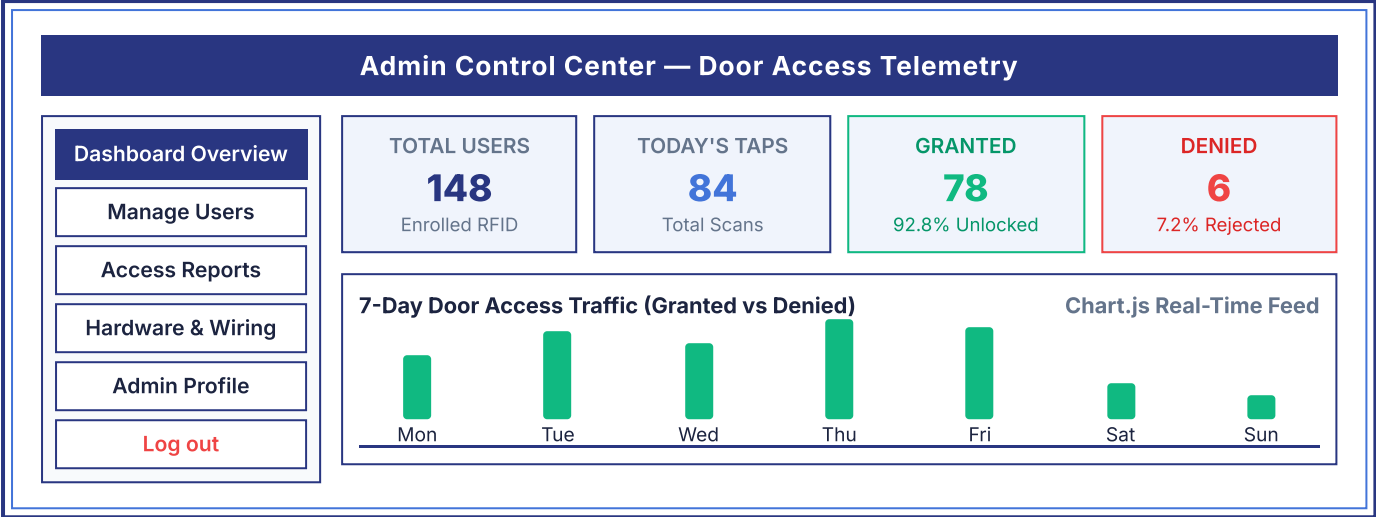
.....

[Go Back to Login Page...](#)

Verify & Reset Password

ADMINISTRATIVE MODULE

DASHBOARD & REAL-TIME MONITORING CENTER



ADMINISTRATIVE MODULE
LIVE ACCESS ACTIVITY & TAP INSPECTOR

Real-Time RFID Access Activity Stream (Door 01 - CICS Main Entrance)

Auto-refreshing via AJAX Polling (2000ms interval)

STREAM ACTIVE •

Pause Feed

Timestamp	Scanned UID	Cardholder Name	Role	Access Result	Hardware Reaction
14:10:05	0A 75 B4 02	Prof. Glenn Delacruz	FACULTY	GRANTED	Relay Unlocked (6s) + MP3 Track 2
14:02:18	FF 20 89 11	Unknown Card (Unregistered)	GUEST	DENIED	Buzzer Alarm + MP3 Track 1
13:45:50	E2 41 89 3F	John Kevin Ramos	STUDENT	GRANTED	Relay Unlocked (6s) + MP3 Track 2
13:30:12	1F 88 A3 90	Carlos Reyes (Account Inactive)	STAFF	DENIED	Buzzer Alarm + MP3 Track 1

Showing last 4 hardware access events.

View Full Access Logs Table



Active Hardware Tap Inspector & Decision Pipeline



ID PHOTO

Prof. Glenn Delacruz

ACCESS GRANTED

Institutional ID: **CICS-FAC-001** | Role: **Faculty (Computer Science)**

Hardware RFID Tag UID: **0A 75 B4 02**

✓ Relay Pin D26 Pulsed LOW for 6000ms | DFPlayer triggered Track 0002.mp3 ("Access Granted, Welcome")

[Inspect Cardholder Profile...](#)

Manual Override / Re-lock

USER MANAGEMENT MODULE
CARDHOLDER DIRECTORY & SECURITY SEARCH

Authorized Cardholder Management (Students, Faculty & Staff)

Search by name, ID number, or RFID UID...

Search

Clear

+ Enroll New User

All Roles (148)

Students (112)

Faculty (24)

Staff (12)

Active Cards (142)

Inactive / Revoked (6)

#	ID Number	Full Name	RFID UID	Role	Status	Action
1	CICS-FAC-001	Prof. Glenn Delacruz	0A 75 B4 02	FACULTY	ACTIVE	Edit Delete
2	2023-0104-CS	Juan Dela Cruz	93 39 6E 1B	STUDENT	ACTIVE	Edit Delete
3	2022-0922-IT	Maria Santos	E2 41 89 3F	STUDENT	ACTIVE	Edit Delete
4	CICS-STF-014	Carlos Reyes	1F 88 A3 90	STAFF	INACTIVE	Edit Delete

Showing 1 to 4 of 148 entries

Previous

1

2

Next



Selected Cardholder Security Summary (RFID Tag #93 39 6E 1B)

Hardware Authorization:

Permanent Whitelist & Offline Emergency Bypass Key

MASTER / ACTIVE

Total Door Taps Logged:

142 Swipes recorded (140 Granted, 2 Denied during maintenance)

[Inspect Taps History...](#)

Temporarily Deactivate Card Access

Generate RFID Card Slip

USER MANAGEMENT MODULE
CARDHOLDER ENROLLMENT WIZARD (STEP 1 & 2)

New Cardholder Enrollment — Step 1: User Identity & Academic Role

1. Identity & Institutional Role

2. Hardware RFID Binding & Tap

Institutional ID Number:

2024-CICS-0192

Full Name:

Juan Dela Cruz

Institutional Role:

Student (BS Information Technology)

Account Status:

Active (Access Allowed immediately upon registration)

ACTIVE

Notice: All cardholder records are synchronized to the local database and cached for fast sub-millisecond door clearance.

Next Step: Bind RFID Card →



New Cardholder Enrollment — Step 2: Hardware RFID Card Binding

1. Identity & Institutional Role

2. Hardware RFID Binding & Tap

RFID Card UID:

0A 75 B4 02

CAPTURED FROM HARDWARE

Live Tap Listener:

Fetch Last Scanned Card from Hardware Reader

Clear Tag

Profile Photo (JPG/PNG):

juan_delacruz_id.jpg (240 KB)

UPLOAD FILE

✓ Scanner Signal Detected: RC522 Reader on COM3 received UID `0A 75 B4 02` at 14:10:05.

Cancel

Complete Registration & Authorize Card

USER MANAGEMENT MODULE
MODIFY PROFILE & ACCESS REVOCATION

Edit Cardholder Profile & Re-assign RFID Keycard

Institutional ID Number:

2023-0104-CS

Full Name:

Juan Dela Cruz

Institutional Role:

Student

Assigned RFID Tag UID:

0A 75 B4 02

Re-Scan New Card

Access Permissions Status:

Active (Authorized to open Solenoid Door Lock)

ACTIVE

Cancel

Save Profile Updates



Revoke Door Access / Delete Cardholder



Confirm Access Revocation for Cardholder?

Are you sure you want to remove **Juan Dela Cruz (2023-0104-CS)**?

Their physical RFID card 0A 75 B4 02 will immediately be blocked and rejected by the door lock.

Revocation Reason:

Graduated / Transferred / Card Lost

Cancel & Keep Active

Set Inactive (Soft Delete - Preserves Logs)

Confirm Permanent Delete

SECURITY AUDIT & REPORTING
ACCESS LOG AUDIT & MULTI-PARAMETRIC FILTER

Door Access History & Multi-Criteria Log Audit

Date Range:2026-09-01 to 2026-09-21

Access Result:All (Granted & Denied)

Role Filter:All Roles (Student, Faculty, Staff)

Search Keyword:Filter by Name, ID, or UID...

Filtered Taps: 1,420

Granted Swipes: 1,350 (95.1%)

Denied / Rejections: 70 (4.9%)

Unregistered Cards: 42

Apply Filters

Reset

Export to CSV

Print Report

Export Official PDF



Verified Security Access Trail (Door 01 - CICS Main Door)						
Log ID	Date & Time	Cardholder / Scanned Identity	RFID UID	Role	Result	Access Reason
#1420	2026-09-21 14:10:05	Prof. Glenn Delacruz	0A 75 B4 02	FACULTY	GRANTED	Authorized Access (OK)
#1419	2026-09-21 14:02:18	Unknown Cardholder	FF 20 89 11	NONE	DENIED	Unknown Card UID
#1418	2026-09-21 13:45:50	Carlos Reyes	1F 88 A3 90	STAFF	DENIED	Inactive User Account
#1417	2026-09-21 13:30:12	Maria Santos	E2 41 89 3F	STUDENT	GRANTED	Authorized Access (OK)
#1416	2026-09-21 12:15:30	System Emergency Key	93 39 6E 1B	ADMIN	GRANTED	Master Card Bypass

Showing 1 to 5 of 1,420 filtered records

« First

< Prev

1

2

3

Next >

SECURITY AUDIT & REPORTING
OFFICIAL ACCESS AUDIT SLIP (FORMAL PRINT VIEW)

CAGAYAN STATE UNIVERSITY — PIAT CAMPUS
College of Information and Computing Sciences (CICS)
SMART DOOR AUTOMATION SYSTEM (SDASFC) ACCESS AUDIT RECORD

Slip Reference No: CSU-SDASFC-2026-0921
Audit Scope: CICS Main Computer Lab Door 01
Reporting Period: 2026-09-01 to 2026-09-21

Date Generated: 2026-09-21 15:30:00
Generated By: Administrator (Engr. Glenn Delacruz)
Integrity Status: **CRYPTOGRAPHICALLY VERIFIED**



Official Summary & Departmental Clearance Sign-Off

TOTAL ATTEMPTS
1,420

CLEARED / UNLOCKED
1,350

SECURITY DENIALS
70

Security Audit Observations & Incident Notes:

- Recorded 42 unauthorized swipes from non-registered RFID keycards (all denied with Track 0001 audio alert).
- Hardware offline emergency master card (93 39 6E 1B) successfully authorized during network test.
- Relay solenoid lock operated normally with zero brownout resets during the audit window.

ENGR. GLENN DELACRUZ
System Administrator / Hardware Engineer

CICS COLLEGE DEAN
College of Information and Computing Sciences

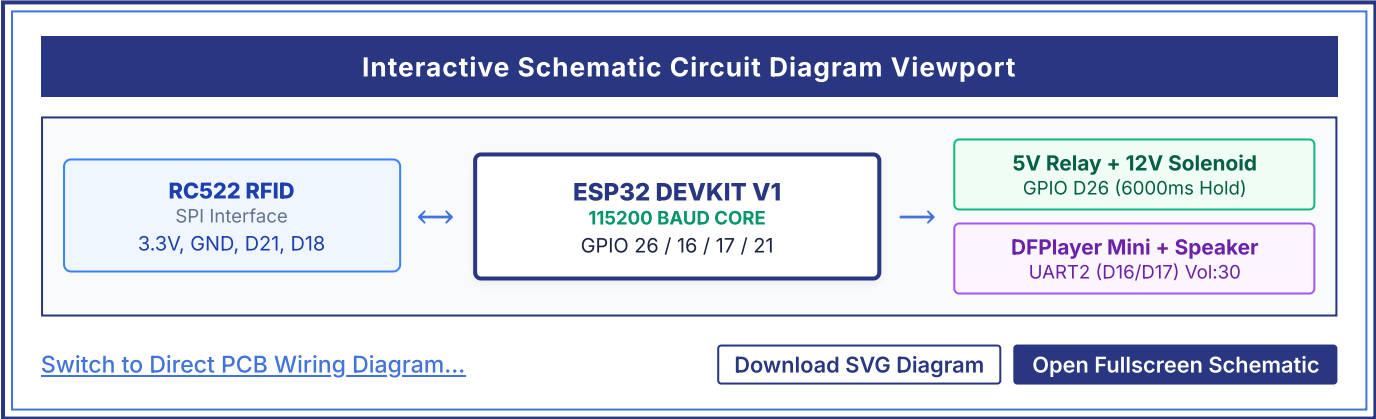
Close Slip View

 Print Formal Slip (A4)

HARDWARE INTEGRATION MODULE
ESP32 CIRCUIT PINOUT & SCHEMATICS VIEWER

Microcontroller Interface & Pin Allocation Matrix			
☒ Breadboard Prototyping Mode		☐ Direct PCB Wiring (Without Breadboard)	
Hardware Subsystem	ESP32 Pinout	Operating Voltage	Operational Specification
RFID-RC522 Scanner	SDA:D21, SCK:D18, MOSI:D23, MISO:D19	3.3V DC (Regulated)	13.56 MHz SPI Reader, reads MIFARE Classic 1K Cards
1-Channel 5V Relay	IN: GPIO D26	5V DC (Coil)	Active-LOW trigger controls 12V Solenoid (6000ms pulse)
DFPlayer Mini MP3	RX: GPIO D17, TX: GPIO D16	5V DC (VCC)	Serial2 UART @ 9600 Baud, 1K resistor on RX pin, Vol: 30
Status Indicator & Buzzer	Buzzer: D4, Green: D2, Red: D15	3.3V / 5V DC	Auditory tone and visual feedback on clearance decision

 Brownout Protection: Dedicated 12V/2A power supply stepped down to 5V via LM2596 buck converter prevents ESP32 reboots during relay inrush.



HARDWARE INTEGRATION MODULE
SERIAL BRIDGE PROTOCOL & OFFLINE FAILSAFE

Serial Bridge Communication Engine (PowerShell / Python / PHP)

Port: COM3 | Baud: 115200 | Buffer: 8-N-1

BRIDGE ONLINE •

```
[BRIDGE 14:10:00] Initialized serial bridge on COM3 at 115200 baud.
[ESP32 14:10:02] SDASFC Firmware v2.4 initialized. Ready for RFID scans.
[ESP32 14:10:05] UID:0A 75 B4 02
[BRIDGE 14:10:05] Captured UID '0A 75 B4 02'. Forwarding HTTP POST to /public/api/rfid_scan.php...
[SERVER 14:10:05] HTTP 200 OK → {"access":"granted","name":"Glenn Delacruz","relay_ms":6000}
[BRIDGE 14:10:05] Transmitting decision command to ESP32: 'GRANT'
[ESP32 14:10:05] COMMAND: GRANT → Relay Pin 26 LOW (6000ms). Audio Track 0002 triggered.
```

Bridges: serial_bridge.ps1, serial_bridge.py, serial_bridge.php

Clear Terminal

Restart Bridge Daemon



Hardware-Level Offline Failsafe & Emergency Master Bypass

Master Key Card UID:

93 39 6E 1B

FIRMWARE HARDCODED

Autonomous Bypass
Logic:

If PC bridge / web server is offline, ESP32 unlocks relay locally.

FAIL-SAFE

Relay Unlock Duration:

6,000 milliseconds (UNLOCK_HOLD_MS = 6000)

6.0s

Diagnostic Simulated Test Controls:

Send 'DENY' Test

Send 'GRANT' Test

ADMINISTRATIVE MODULE
ADMINISTRATOR CREDENTIALS & SECURITY PROFILE

Administrator Profile & Security Challenge Configuration

Username:

admin

SYSTEM ROOT

Full Name:

Engr. Glenn Delacruz

Configured Security Question:

What is the primary server room designation?

Security Answer:

..... (Hash stored via Bcrypt)

ENCRYPTED

Save Profile & Security Question



Update Master Password & Session Security

Current Password:

.....

New Password:

.....

Confirm New Password:

.....

✓ Minimum 8 characters

✓ Uppercase & lowercase

✓ Numbers & special symbols

Log Out of All Active Sessions

Log Out of System

Update Password

SPECIFICATION & DESIGN SYSTEM

SYSTEM ARCHITECTURE & UNIFORM COLOR PALETTE

End-to-End Hardware & Software Access Control Pipeline

- STEP 1:** Cardholder taps physical RFID card on **RC522 Scanner** (SPI Interface, 13.56 MHz).
- STEP 2:** **ESP32 Microcontroller** captures UID, checks Master Key bypass, then sends UID: <HEX> over USB Serial (115200 baud).
- STEP 3:** **Serial Bridge Daemon** receives serial line and issues HTTP POST JSON to /public/api/rfid_scan.php.
- STEP 4:** **PHP AccessController** queries MySQL users table, checks status (Active/Inactive), logs tap into access_logs.
- STEP 5:** Bridge sends GRANT or DENY to ESP32. Relay unlocks solenoid for **6000ms**; DFPlayer plays Track 2 or 1.



Uniform Color System & Component Design Specifications

					
CICS Navy #293681	Accent Blue #4274d9	Sky Light #95ccdd	Granted #10b981	Denied #ef4444	Slate Ink #1c2440

Uniformity Directives: All screens follow the dual-line wireframe container convention. High-priority interactive buttons strictly use CICS Navy (#293681) or Accent Blue (#4274d9). Positive validations and door unlocks employ Emerald Green (#10b981). Lockout alerts and security denials employ Coral Red (#ef4444). Monospaced code styling applies uniformly to all RFID UIDs, ports, and hardware parameters.

Cagayan State University — College of Information and Computing Sciences

[Print Complete 12-Page Prototype Document \(PDF\)](#)